

(residual in residual dense block), and a local residual fusion unit (LRFU). The RRDB module can improve the information flow and gradients, while the LRFU module can effectively fuse multi-level residual information in the local fusion block. The model was trained using VGG-based, SSIM, and Smooth L_1 losses.

4.8. CVML

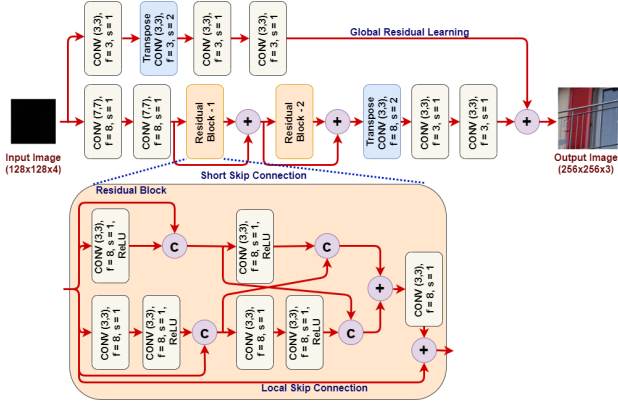


Figure 10. CNN architecture proposed by CVML team.

Figure 10 demonstrates the model proposed by team CVML. This architecture is using residual blocks to extract a rich set of features from the input data. Transposed convolution layer is used to upsample the final feature maps to the target resolution. To stabilize the training process, a global residual learning strategy was employed that also helped to reduce the color shift effect. The model was trained to minimize the combination of L_1 and SSIM losses and was optimized using Adam with a learning rate of $1e-4$ for 1M iterations.

4.9. EdS

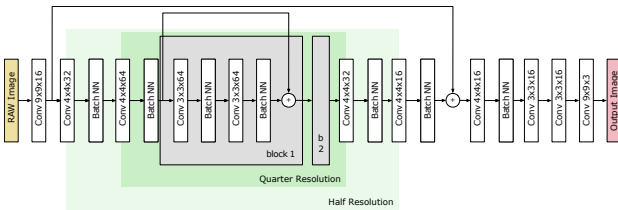


Figure 11. Residual network proposed by team EdS.

EdS proposed a ResNet-based architecture shown in Fig. 11 that was derived from [16]. The main difference consists in using two 4×4 convolutional layers with stride 2 for going into lower-dimensional space, and additional skip connections for faster training. The network was trained for 33K iterations using the same losses and setup as in [8].

5. Additional Literature

An overview of the past challenges on mobile-related tasks together with the proposed solutions can be found in the following papers:

- Learned End-to-End ISP: [25, 29]
- Perceptual Image Enhancement: [28, 22]
- Bokeh Effect Rendering: [21, 27]
- Image Super-Resolution: [28, 41, 4, 48]

Acknowledgements

We thank MediaTek Inc., AI Witchlabs and [ETH Zurich](#) (Computer Vision Lab), the organizers and sponsors of this Mobile AI 2021 challenge.

A. Teams and Affiliations

Mobile AI 2021 Team

Title:

Mobile AI 2021 Learned Smartphone ISP Challenge

Members:

Andrey Ignatov^{1,3} (andrey@vision.ee.ethz.ch), Cheng-Ming Chiang² (jimmy.chiang@mediatek.com), Hsien-Kai Kuo² (hsienkai.kuo@mediatek.com), Anastasia Sycheva¹, Radu Timofte^{1,3} (radu.timofte@vision.ee.ethz.ch), Min-Hung Chen² (mh.chen@mediatek.com), Man-Yu Lee² (my.lee@mediatek.com), Yu-Syuan Xu² (yu-syuan.xu@mediatek.com), Yu Tseng² (ulia.tseng@mediatek.com)

Affiliations:

¹ Computer Vision Lab, [ETH Zurich](#) Switzerland

² MediaTek Inc., Taiwan

³ AI Witchlabs, Switzerland

dh_isp

Title:

Smallnet for An End-to-end ISP Pipeline

Members:

Shusong Xu (13821177832@163.com), Jin Guo

Affiliations:

Dahua Technology, China

AIISP

Title:

CSAnet: High Speed Channel Spatial Attention Network for Mobile ISP [12]

Members:

Chao-Hung Chen (ZHChen@itri.org.tw), Ming-Chun